

=== GNB (1780954758) ===

This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.

This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 10 MHz bandwidth.

To run the srsRAN Project gNB with this config, use the following command:

sudo ./gnb -c gnb_zmq.yaml

cu_cp:

amf:

addr: 10.53.1.2 # The address or hostname of the AMF.

port: 38412

bind_addr: 10.53.1.1 # A local IP that the gNB binds to for traffic from the AMF.

supported_tracking_areas:

- tac: 7

plmn_list:

- plmn: "00101"

tai_slice_support_list:

- sst: 1 # 3GPP TS 23.501 Table 5.15.2.2-1: eMBB

inactivity_timer: 2700 # Sets the UE/PDU Session/DRB inactivity timer to 2700 seconds (45 minutes). Supported: [1 - 7200].

ru_sdr:

device_driver: zmq # The RF driver name.

device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass arguments to the selected RF driver.

srate: 11.52 # RF sample rate might need to be adjusted according to selected bandwidth.

tx_gain: 75 # Transmit gain of the RF might need to adjusted to the given situation.

rx_gain: 75 # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:

dl_arfcn: 368500 # ARFCN of the downlink carrier (center frequency).

band: 3 # The NR band.

channel_bandwidth_MHz: 10 # Bandwidth in MHz. Number of PRBs will be automatically derived.

common_scs: 15 # Subcarrier spacing in kHz used for data.

plmn: "00101" # PLMN broadcasted by the gNB.

tac: 7 # Tracking area code (needs to match the core configuration).

pdccch:

common:

ss0_index: 0 # Set search space zero index to match srsUE capabilities

coreset0_index: 6 # Set search CORESET Zero index to match srsUE capabilities

dedicated:

ss2_type: common # Search Space type, has to be set to common

dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)

prach:

prach_config_index: 1 # Sets PRACH config to match what is expected by srsUE

pdsch:

mcs_table: qam64 # Sets PDSCH MCS to 64 QAM

pusch:

mcs_table: qam64 # Sets PUSCH MCS to 64 QAM

log:

filename: /tmp/gnb.log # Path of the log file.

all_level: info # Logging level applied to all layers.

hex_max_size: 0

pcap:

mac_enable: false # Set to true to enable MAC-layer PCAPs.

mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.

ngap_enable: true # Set to true to enable NGAP PCAPs.

ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.

e2ap_enable: true # Set to true to enable E2AP PCAPs.

e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.

e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.

e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

```
#e2:
#  enable_du_e2: true           # Enable DU E2 agent (one for each DU instance)
#  e2sm_kpm_enabled: true      # Enable KPM service module
#  e2sm_rc_enabled: true       # Enable RC service module
#  addr: 127.0.0.1             # RIC IP address
#  bind_addr: 127.0.0.100      # A local IP that the E2 agent binds to for traffic from the RIC. ONLY
required if running the RIC on a separate machine.
#  port: 36421                 # RIC port

metrics:
  layers:
    enable_rlc: true           # Enable RLC metrics
    enable_sched: true         # Enable DU scheduler metrics
  periodicity:
    du_report_period: 400      # Set DU statistics report period to 400ms
```

=== UE (1780954758) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 3
nof_carriers = 1

max_nof_prb = 52
nof_prb = 52

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 112233445566778899aabbccddeeff00
k = 00aabbccddeeff112233445566778899
imsi = 0010100000000001
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1780954758) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
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#
# A copy of the GNU Affero General Public License can be found in
# the LICENSE file in the top-level directory of this distribution
# and at http://www.gnu.org/licenses/.
#
```

services:

5gc:

```
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  environment:
    - MONGODB_IP=127.0.0.1
    - OPEN5GS_IP=10.53.1.2
    - UE_IP_BASE=10.45.0
    - UPF_ADVERTISE_IP=10.53.1.2
    - DEBUG=false
```

SUBSCRIBER_DB=001010000000001,00aabbccddeeff112233445566778899,opc,112233445566778899aabbccddeeff00,8000,9,10.45.1.2

```
- NETWORK_NAME_FULL=srsRAN
- NETWORK_NAME_SHORT=srsRAN
- TZ=Europe/Madrid
```

privileged: true

ports:

```
- "9898:9999/tcp"
```

Uncomment port to use the 5gc from outside the docker network

```
# - "38412:38412/sctp"
```

```
# - "2152:2152/udp"
```

command: 5gc -c open5gs-5gc.yml

healthcheck:

```
  test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
```

interval: 3s

timeout: 1s

retries: 60

networks:

ran:

```
  ipv4_address: ${OPEN5GS_IP:-10.53.1.2}
```

gnb:

container_name: srsran_gnb

Build info

image: srsran/gnb

build:

context: ..

dockerfile: docker/Dockerfile

```

args:
  OS_VERSION: "24.04"
# privileged mode is required only for accessing usb devices
privileged: true
# Extra capabilities always required
cap_add:
  - SYS_NICE
  - CAP_SYS_PTRACE
volumes:
  # Access USB to use some SDRs
  - /dev/bus/usb/:/dev/bus/usb/
  # Sharing images between the host and the pod.
  # It's also possible to download the images inside the pod
  - /usr/share/uhd/images:/usr/share/uhd/images
  # Save logs and more into gnb-storage
  - gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current netowrk configuration creastes a private netwoek with both containers attached
# An alterantive would be `network: host`. That would expose your host network into the container. It's
the easiest to use if the 5gc is not in your PC
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.1}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
gnb_config.yml:
  file: ${GNB_CONFIG_PATH:-./gnb_zmq.yaml} # Path to your desired config file
gnb_compose_config.yml:
  content: |
    cu_cp:
      amf:
        addr: ${OPEN5GS_IP:-10.53.1.2}
        bind_addr: 0.0.0.0
    metrics:
      autostart_stdout_metrics: true
      enable_json: true
    remote_control:
      bind_addr: 0.0.0.0
      enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:
      driver: default
      config:
        - subnet: 172.19.1.0/24

```

=== NOTAS (1780954758) ===

- Se ha aplicado la Regla 1 (Coherencia PLMN) y Regla 7 (Coherencia IMSI y SUBSCRIBER_DB) para MCC 001, MNC 01 y TAC 7 en gNB, UE y Docker.
- Se ha verificado la Regla 2 (Coherencia de frecuencias) para Banda 3 y dl_arfcn 368500.
- Se ha verificado la Regla 6b (Coherencia srates/ancho de banda) para srates 11.52 MHz y ancho de banda 10 MHz.
- Se ha aplicado la Regla 4 (Puertos ZMQ cruzados) para los puertos 2000 y 2001.
- Se ha aplicado la Regla 5 (bind_addr) para gNB standalone (10.53.1.1) y docker-compose (0.0.0.0).
- Se ha copiado el bloque ru_sdr del CAG sin modificaciones, según la Regla 6a.
- Se ha configurado el SST a 1 para eMBB en el gNB, referenciando 3GPP TS 23.501 Table 5.15.2.2-1.
- Se ha configurado el inactivity_timer del gNB a 2700 segundos (45 minutos).
- Se han activado las métricas RLC y del scheduler DU con un periodo de reporte de 400 ms en el gNB.
- Se han habilitado las capturas PCAP de NGAP en el gNB.
- Se ha ajustado nof_prb en el UE a 52 para un ancho de banda de 10 MHz con 15 kHz SCS.
- La Regla 3 (Coherencia SCS) se ha interpretado como aplicable solo si ambos campos (common_scs y ssb_scs) están presentes en la plantilla. Dado que ssb_scs no está en la plantilla gnb_zmq.yaml, no se ha añadido para evitar posibles errores de parsing.